University Chatbot

Hybrid approach: Elastic Search + Retrieval Augmented Generation

Group 7

Motivation

- The Data Labyrinth: Universities operate as massive, decentralized data mazes. Critical academic information is spanning strict tuition policies to unforgiving registration deadlines is buried deeply across dozens of disconnected websites, departmental portals, and hundreds of pages of dense PDF handbooks.
- Paralyzing Friction: This severe fragmentation creates immense, unnecessary friction in routine academic decision-making.
- Students are forced to spend countless hours frantically navigating several unrelated pages just to find a single definitive answer, rather than focusing on their actual coursework.
- Dangerous Consequences: Navigating these chaotic and often outdated portals isn't just stressful—it is actively dangerous. It drastically increases the risk of students acting on misremembered or superseded information, which can lead to disastrous financial penalties, missed deadlines, or derailed graduation timelines.

What is…

- **RAG** - A technique where an LLM is paired with a retrieval system. Instead of relying solely on training knowledge, the model fetches relevant documents at query time and uses them as context to generate accurate, grounded answers.

- **Pipeline**
  Embed query → semantic search over a vector index → pass top-k chunks to the LLM→ generate

- **Hybrid RAG -** is a method that combines exact-match keyword search (BM25) with vector search. It fuses these two ranked lists together, reranks the combined results, and then generates a response.

- **Pipeline**
  BM25 ⨁ kNN → Reciprocal Rank Fusion → cross-encoder rerank → grounded LLM answer

Comparison

Dimension	Vanilla RAG	RAG + Elasticsearch
Retrieval Method	Semantic (vector only)	Hybrid: BM25 + kNN, fused via RRF
Keyword Precision	Poor, misses exact matches	Strong, BM25 handles exact terms
Scalability	Limited by vector store	Horizontally scalable
Filtering	None	Native structured filters
Result Quality	Good for semantic similarity	Better overall, catches what semantic miss

Elastic Search

A distributed search engine built on Apache Lucene. Supports full-text BM25 keyword search and dense vector (kNN) search in the same index. Designed for low-latency retrieval over large structured and unstructured datasets.

Why Elasticsearch

- Lets us run BM25 and kNN side-by-side on the same index, with structured pre-filters on fields like school, level, and term so we never retrieve fee records from the wrong programmed.

Why ES inside RAG

- Pure vector retrieval blursentity specific tokens (school names, fee categories, dates). Adding BM25 restores exact-match precision, while semantic search keeps recall on conversational phrasing.

Why over Vanilla RAG

- Vanilla RAG retrieves semantically similar but factually off-target chunks. ES+ RAG also reranks fused results with a cross-encoder, sending higher quality context to the LLM.

Data Sources

- The system is built on data sources from IIT websites including policies, tuition fees, contacts directory, and academic calendar.
- This comes up to over 44 html links and 2 student hand book pdfs.
- We divided this into two forms based on the format of the data: Structured and Unstructured data.
- Structured data is all of academic calendar, tuition and fees, and contacts directory.
- Unstructured data is all the academic policies

Data Sources

Structured

- Tuition fee tables (HTML)
- Academic calendar events
- Staff & faculty directory (1,552 records)
- Fee schedules (5 schools, 3 levels)
- Row-based — each row = 1 document

Unstructured

- Student Handbook PDF (188 pages)
- Co-terminal handbook (13 pages)
- Policy pages (HTML prose)
- Commencement, Coursera pages
- Registration & withdrawal policies

Chunking

- What is chunking?

Breaking long documents into smaller segments that preserve meaning and fit within retrieval windows.

Strategies:

1. Topic + Sentence

2. Topic + Fixed

3. Topic + Recursive

4. Topic + Semantic

Selected: Hybrid Topic + Sentence

- Divides at section headings (topic boundaries)
- Splits further at sentence boundaries
- No overlap needed — chunks stay coherent
- Matched Semantic quality with zero GPU overhead
- Min token threshold added to merge short chunks

Structured Data — No Chunking Needed

- Calendar events, tuition rows, contacts
- Each record is already a self-contained unit
- Indexed directly with structured field schemas

Data Challenges

Tabular Data Conversion

HTML fee & calendar tables required column header mapping to JSON. Merged cells and inconsistent formatting caused data loss. Required careful parsing per school page.

Chunk Size Calibration

Short chunks (<50 tokens) lacked context. Long chunks (400+ tokens) buried answers. Multiple draft iterations introduced a minimum token threshold and sentence-boundary preservation.

Missing / Heterogeneous Fields

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Noisy Unstructured Docs

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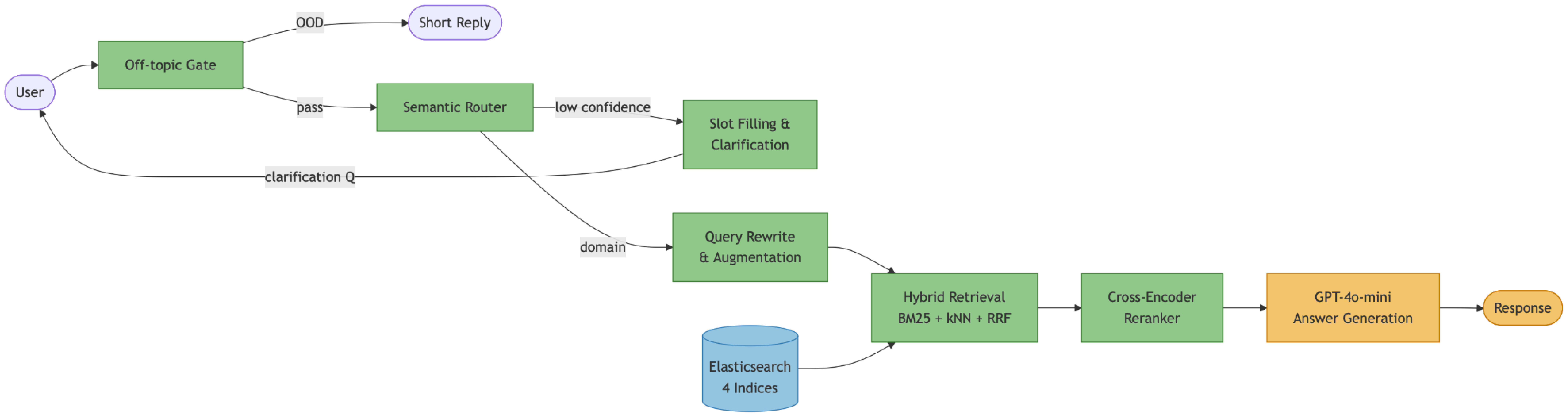
Metadata Formation

Short chunks (<50 tokens) lacked context. Long chunks (400+ tokens) buried answers. Multiple draft iterations introduced a minimum token threshold and sentence-boundary preservation.

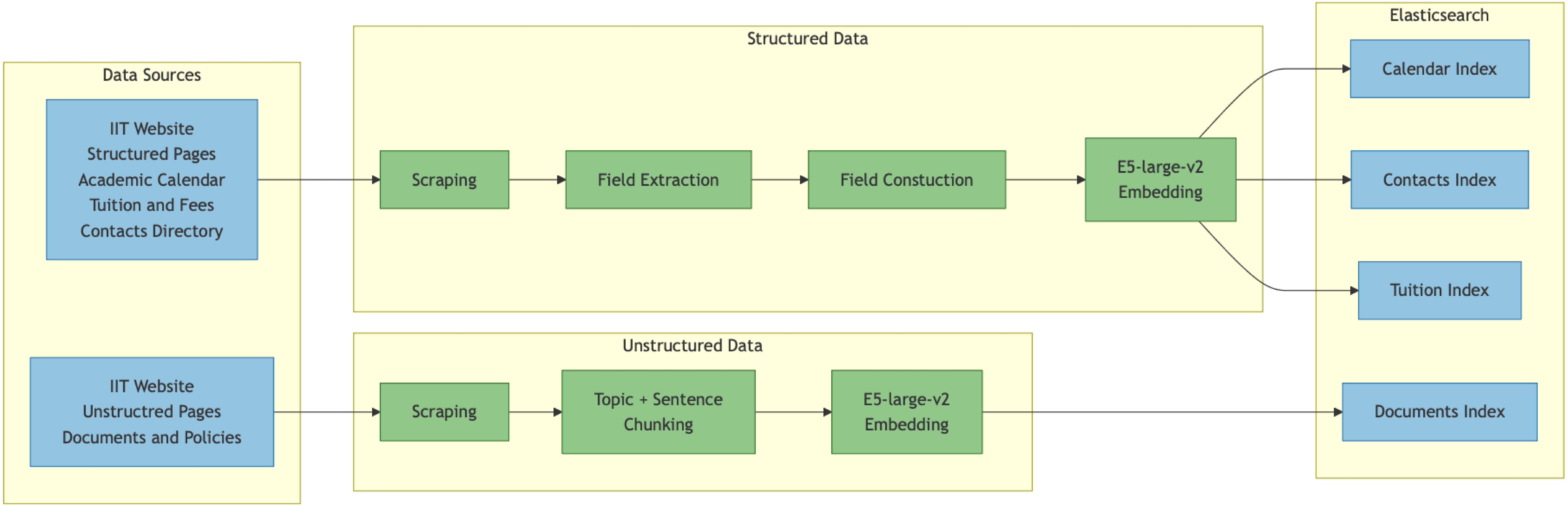
Outdated Content

Retrieval cannot distinguish recency from relevance. Manual review excluded superseded pages (old academic years) before indexing.

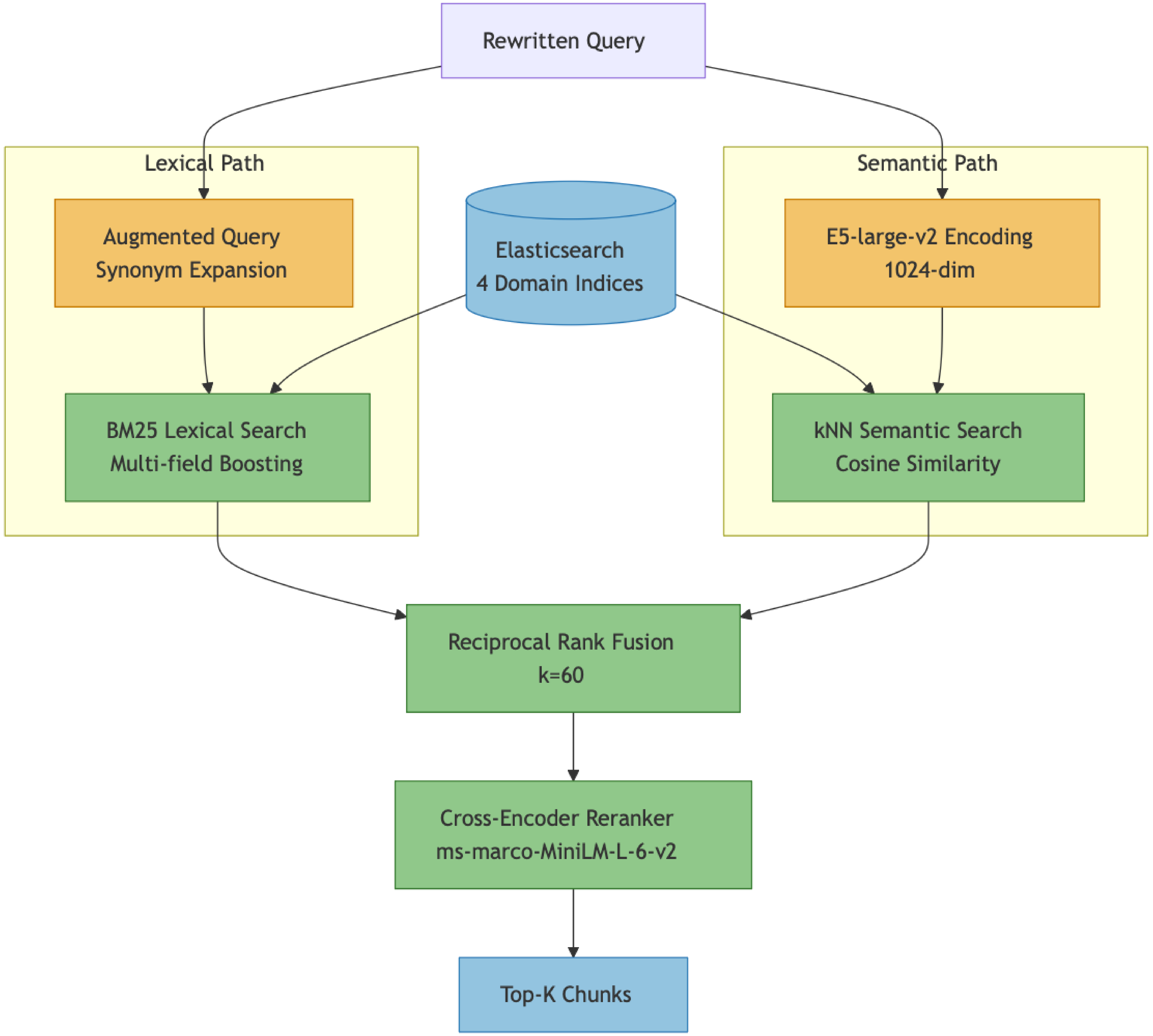
System Design



Ingestion Layer



Retrieval Layer



Key Components

Semantic Routing

- Prototype-based similarity — no labelled data required
- Domain prototype questions curated per domain
- Prototypes precomputed at startup with E5-large-v2
- Query embedded with 'query: ' prefix → cosine vs prototypes
- Confidence threshold: 0.55
- Multi-domain: secondary domain included if within 75% of top
- Failed: keyword routing (natural language), LLM routing (slow/inconsistent)

Slot filling and clarification

- Needs school + level (15 combos) - asks which program
- Needs term/year for event queries - date-exact passes through
- Needs dept - 17 common offices shown as options
- Needs policy topic - long queries (5+ words) pass through

Key Components

Query Rewriting (Semantic Path)

- Uses LLaMA 3.1-8B via Groq API
- Converts casual/pronoun-heavy queries → formal academic vocabulary
- Example: 'what about grad students?' → 'graduate student tuition fees'
- Prior user turn passed as context hint for co-reference resolution
- Rewritten query used for semantic (kNN) path
- Original query retained separately for reranking

Query Augmentation (Lexical Path)

- Domain-specific synonym dictionaries per domain
- Appends synonyms to query: expanded query for BM25
- Dictionaries are intentionally narrow (noise reduction)
- NOT applied to semantic search, embeddings already handle paraphrasing
- Documents domain: expanded query also passed to reranker
- Example: 'drop' → 'withdraw, course removal'
- Improves lexical recall without distorting semantic signal

Key Components

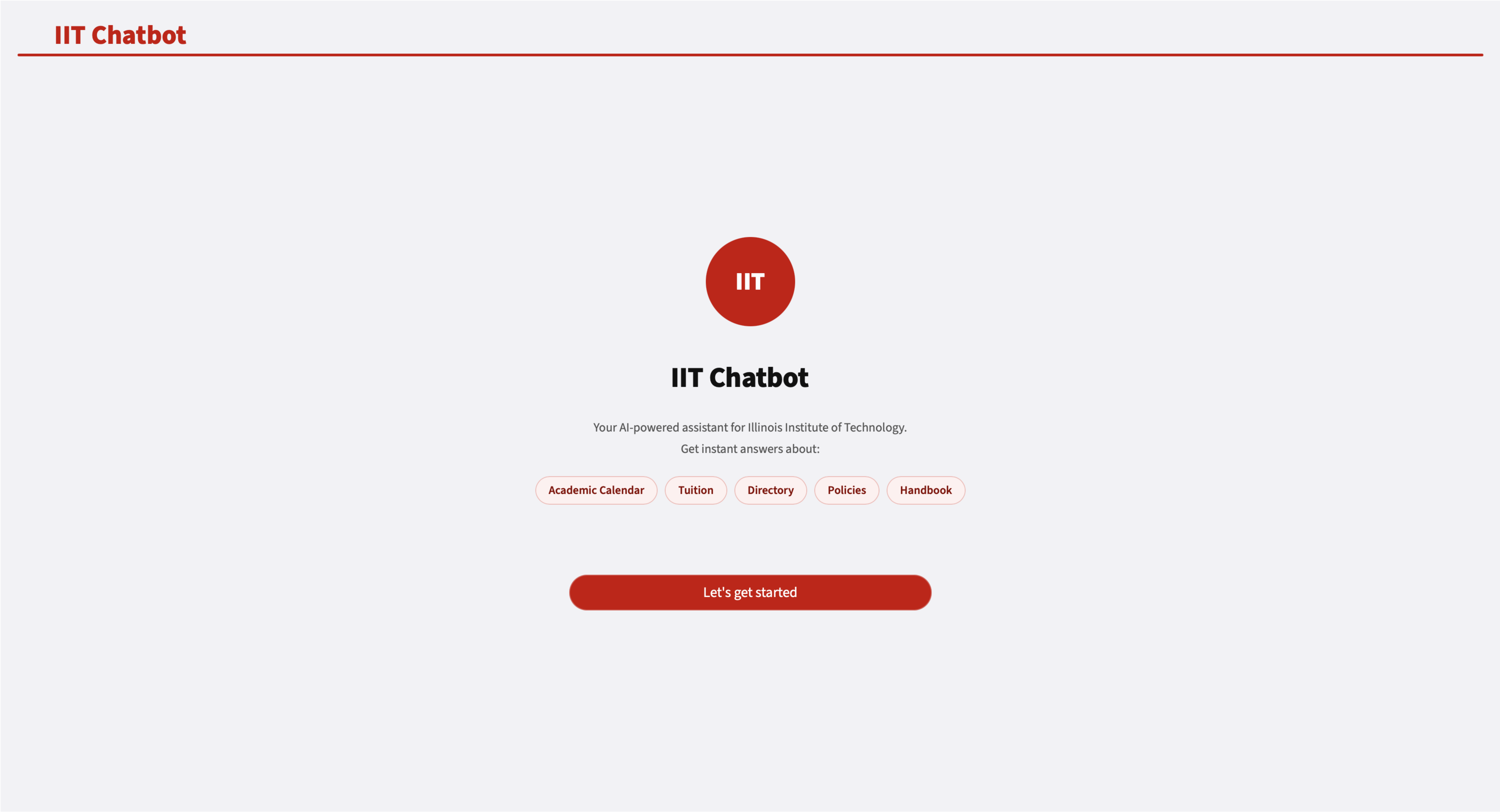
- LLM Generation: GPT-4o

- GPT-4o via Azure OpenAI — primary generation
- LLaMA 3.1-8B via Groq — fallback if Azure unavailable
- Strict grounding: answer ONLY from retrieved context
- Responds 'I don't have that information' if context insufficient
- Exact figures (fees, dates, GPAs) stated verbatim from context
- No conversational filler ('Great question!', pleasantries)
- Multi-level tuition: lists all levels if user hasn't specified
- Conversation history for co-reference only — no prior answer bleed

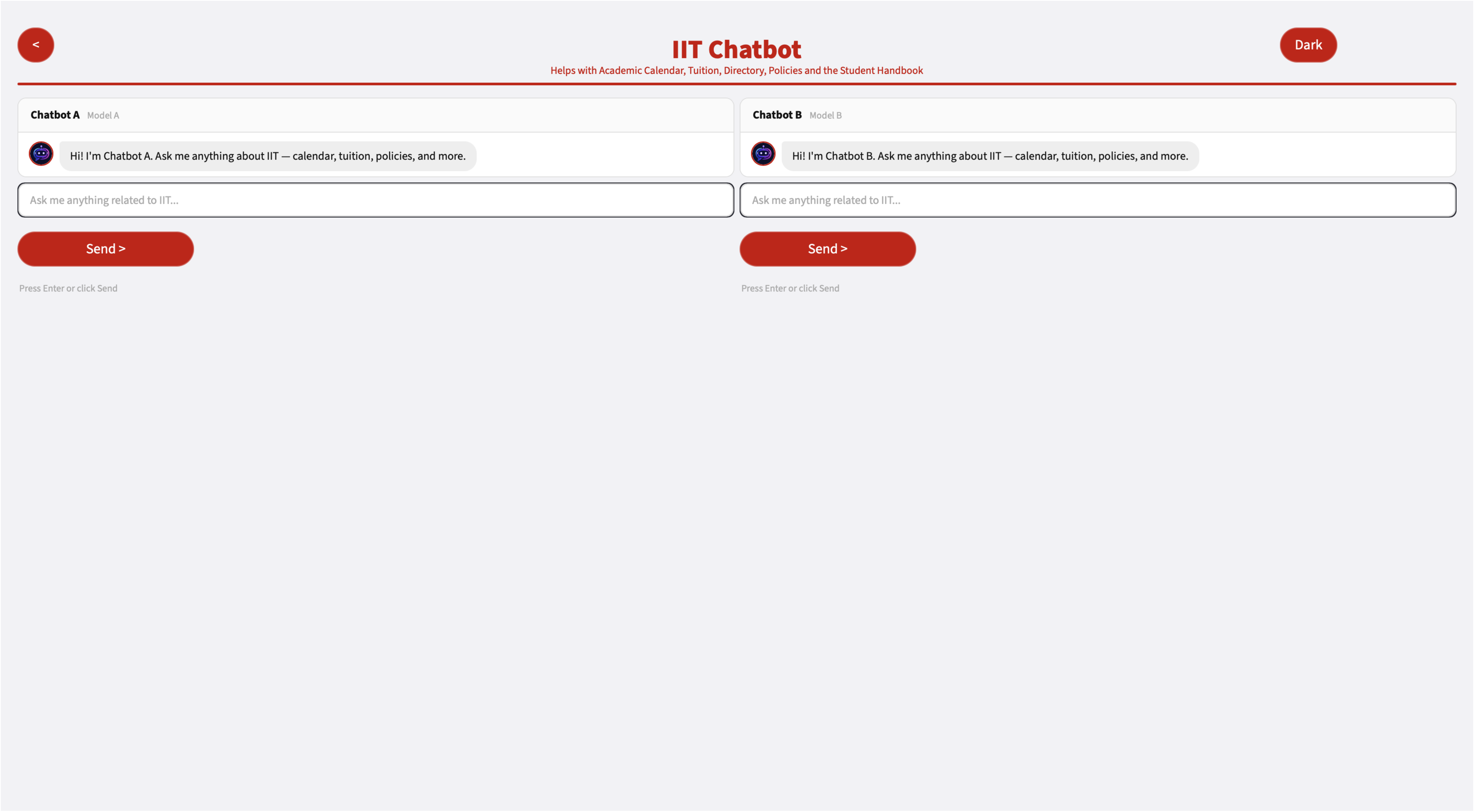
Cross-Encoder Reranking

- Binary encoder - Encodes query & doc independently, fast but misses term interactions
- Cross encoder - Scores query+doc together as one input, captures relevance signals bi-encoder misses
- Model: ms-marco-MiniLM-L-6-v2 (MS MARCO benchmark)
- Topic-match boost proportional to query content words in chunk topic field
- Top-k cap: 5 (calendar/contacts/tuition), 7 (documents), 10 (tuition diversification)

User Interface



User Interface



Technical Challenges in UI

- Streamlit session loss on reload

Conversation history disappeared with a browser refresh. Backend was stateless by design, so no server-side recovery.

- Concurrent users on shared instance

Streamlit state is per-tab, but FastAPI and Elasticsearch are shared — required

- Azure latency

Three network hops (Streamlit → FastAPI → Azure OpenAI) pushed end-to-end response to 3-6 s under normal load.

- Token-limit truncation

Long retrieved chunks occasionally got cut off before reaching the model — required tighter chunk-size targets.

Evaluation

Evaluation is carried out in three stages:

1. Within group evaluation

2. Cross group evaluation

3. Online evaluation

Within group evaluation

- This is done across multiple dimensions and was assessed as “PASS”, “FAIL” OR “PARTIAL”.
- Dimensions:
  - Router Accuracy - The intent router was evaluated on 25 queries with the Pass percentage of 68%
  - Single Turn Query Evaluation - A set of 60 questions was evaluated across 5 domains including out-of-scope with a 80% pass rate
  - Multi turn was evaluated across 30 scenarios testing - clarification, follow up and edge cases achieving an overall 63% pass rate
  - Blind test was evaluated across 42 newer queries and it has a 57% pass rate.

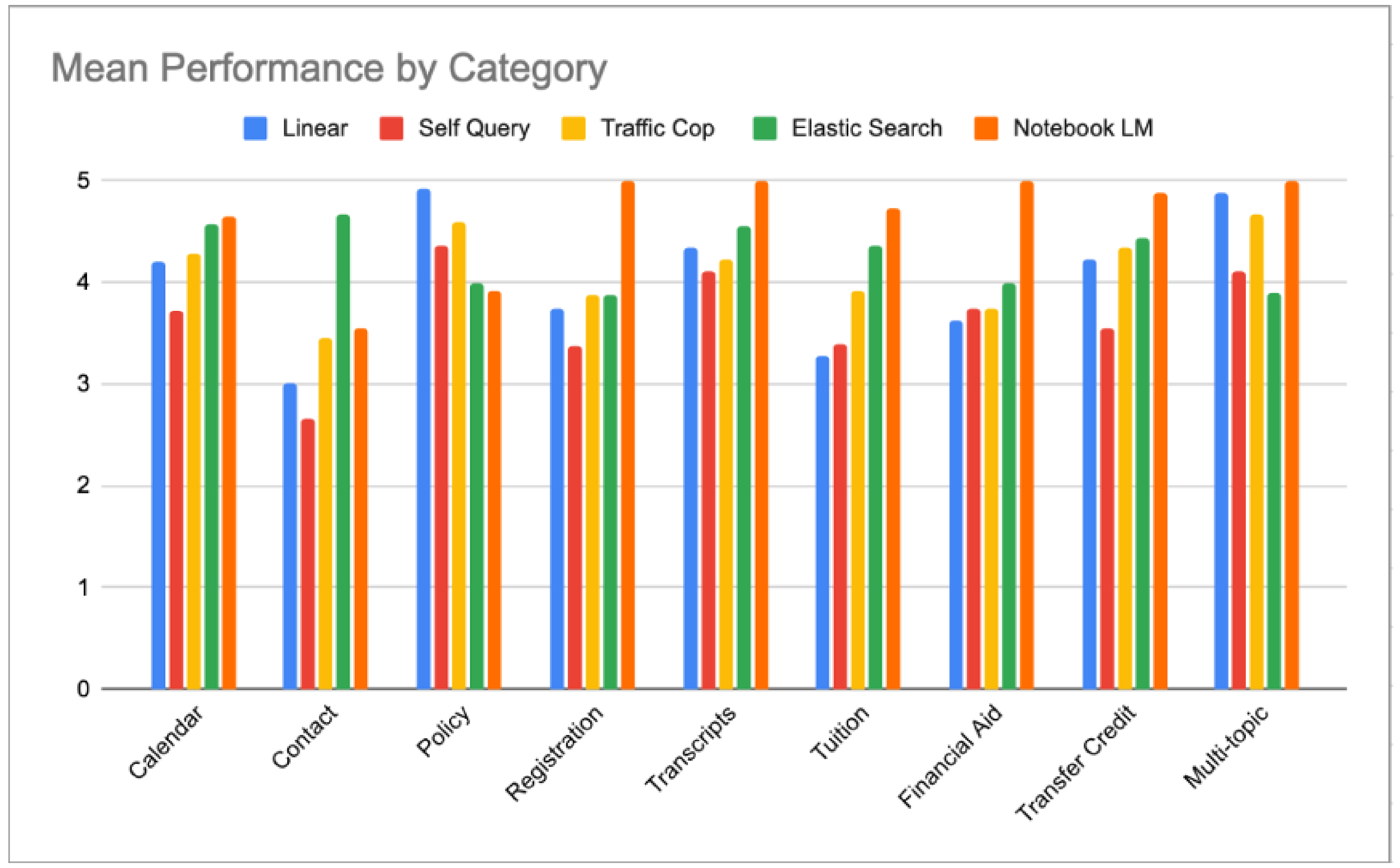
Cross Group Evaluation

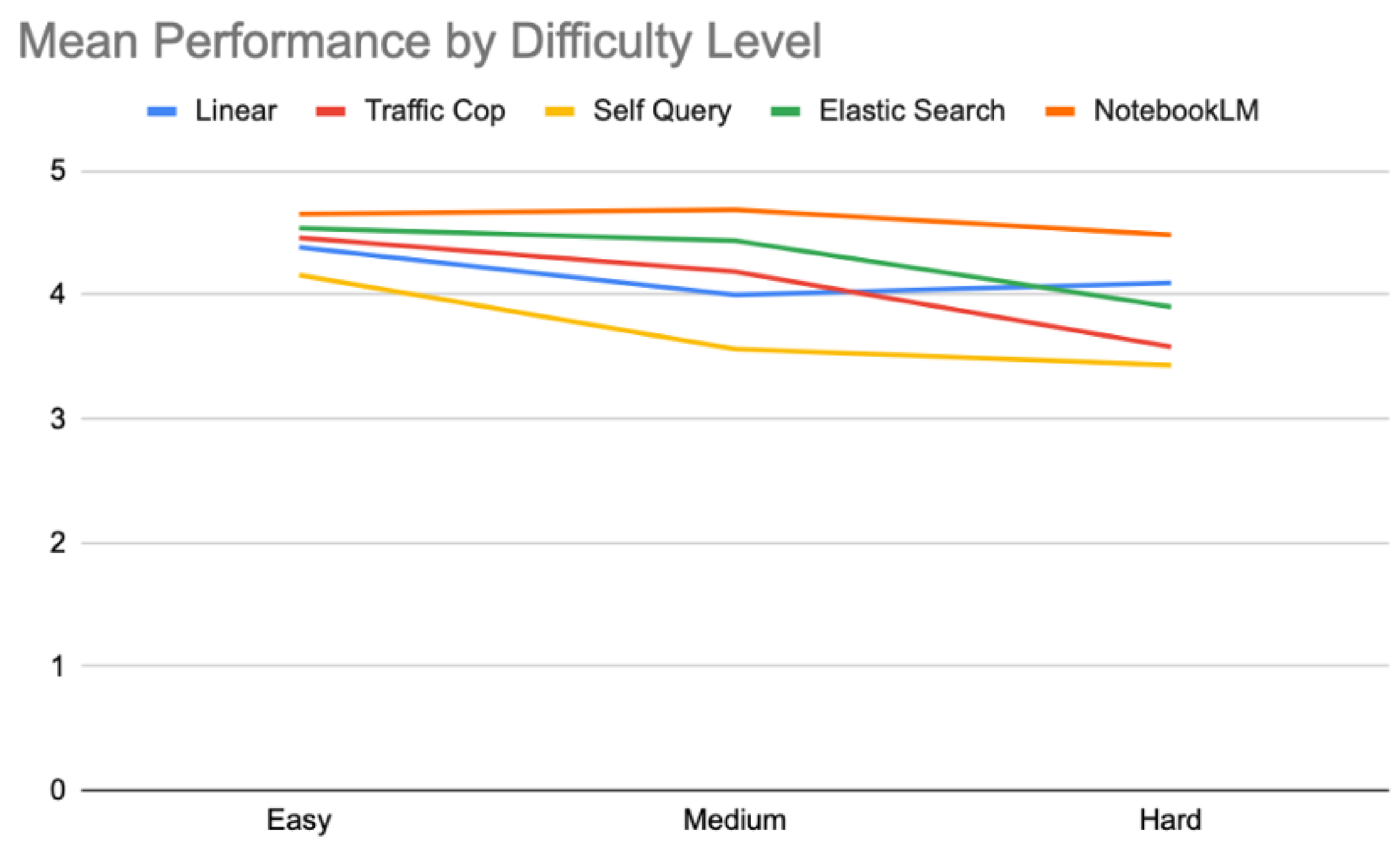
Evaluation categories

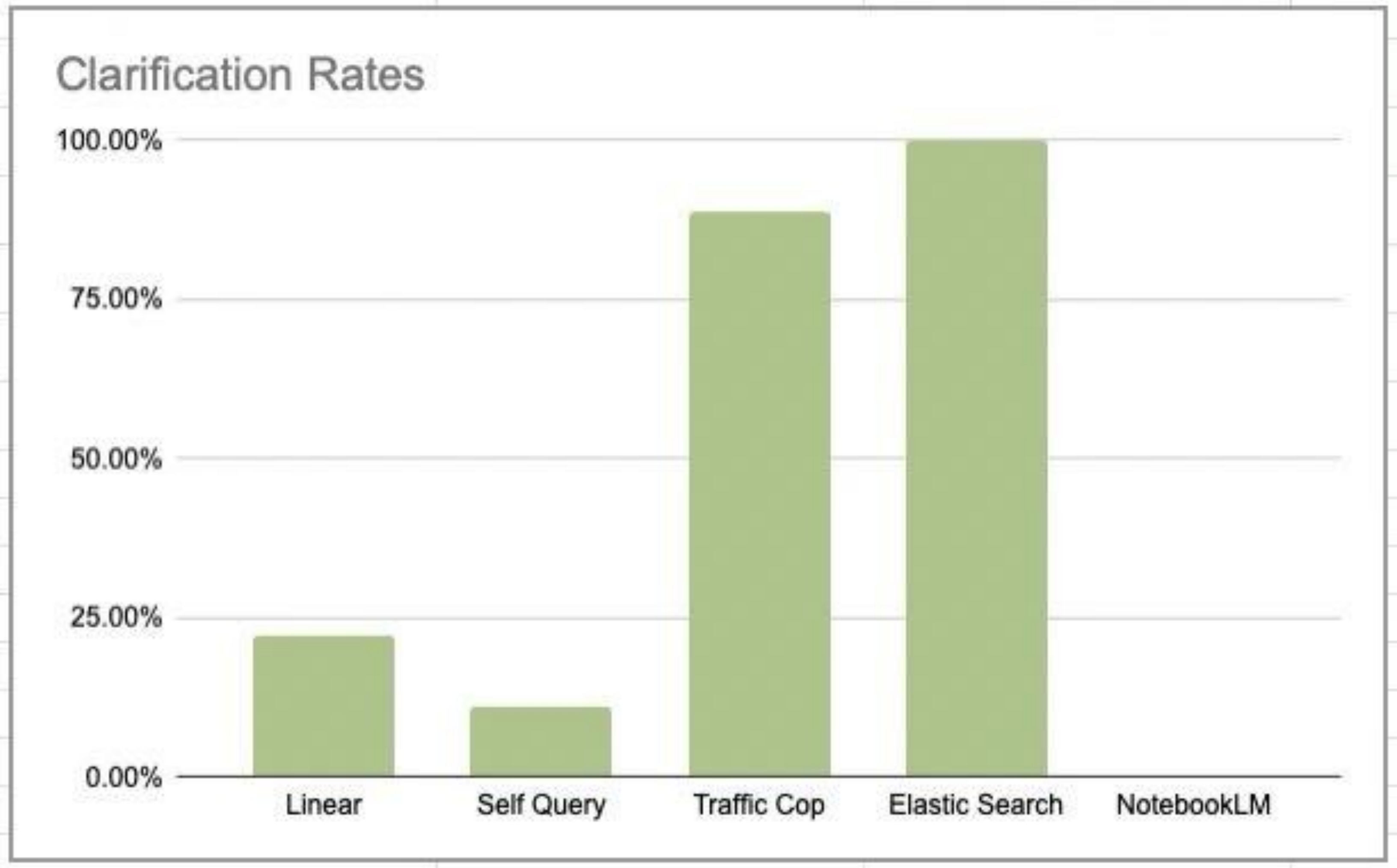
For this evaluation 88 questions across multiple categories were tested on different difficulty levels.

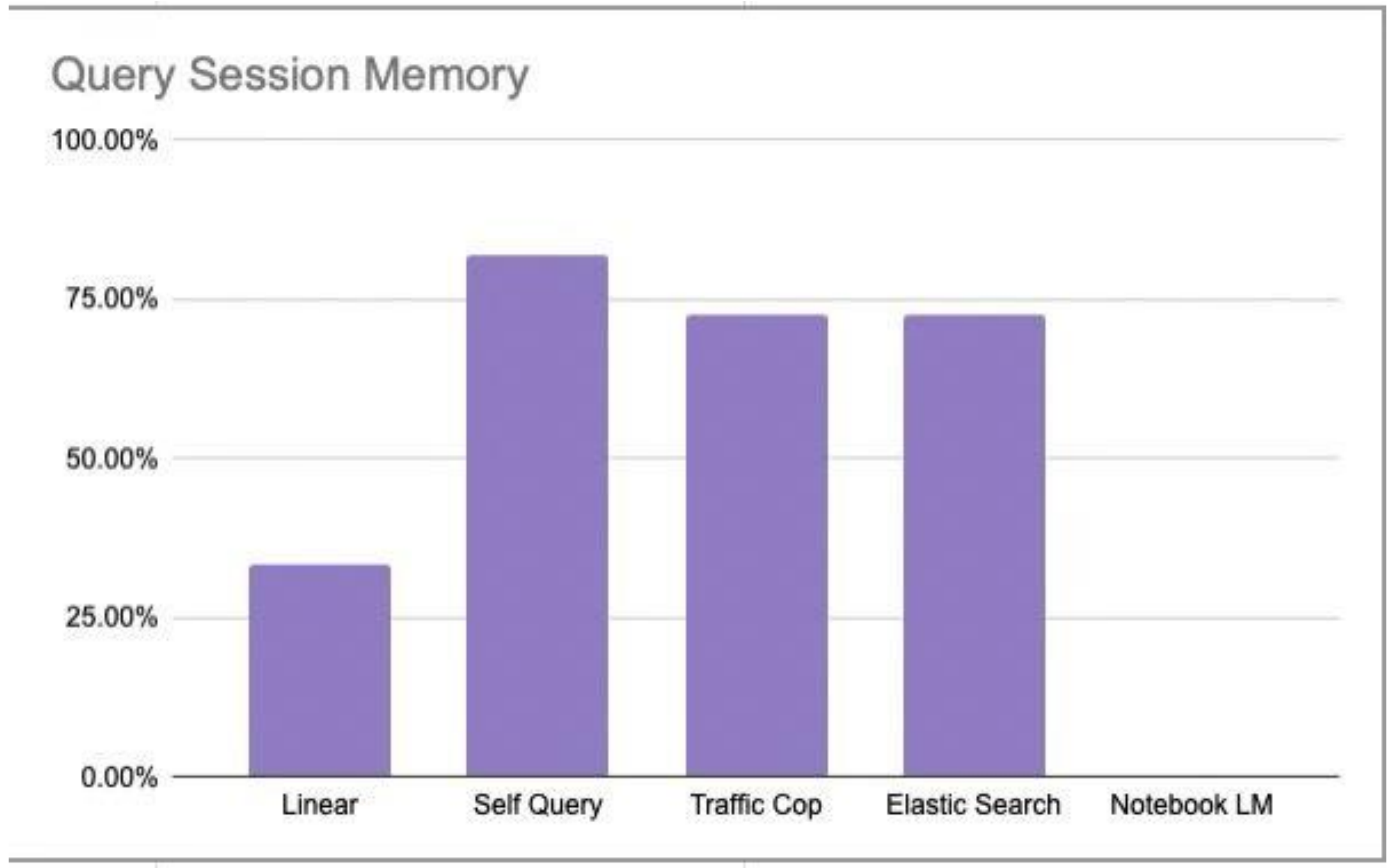
- performance by category
- performance by level
- failure rate analysis
- stress questions
- clarification questions
- query session memory questions

Cross Group Evaluation









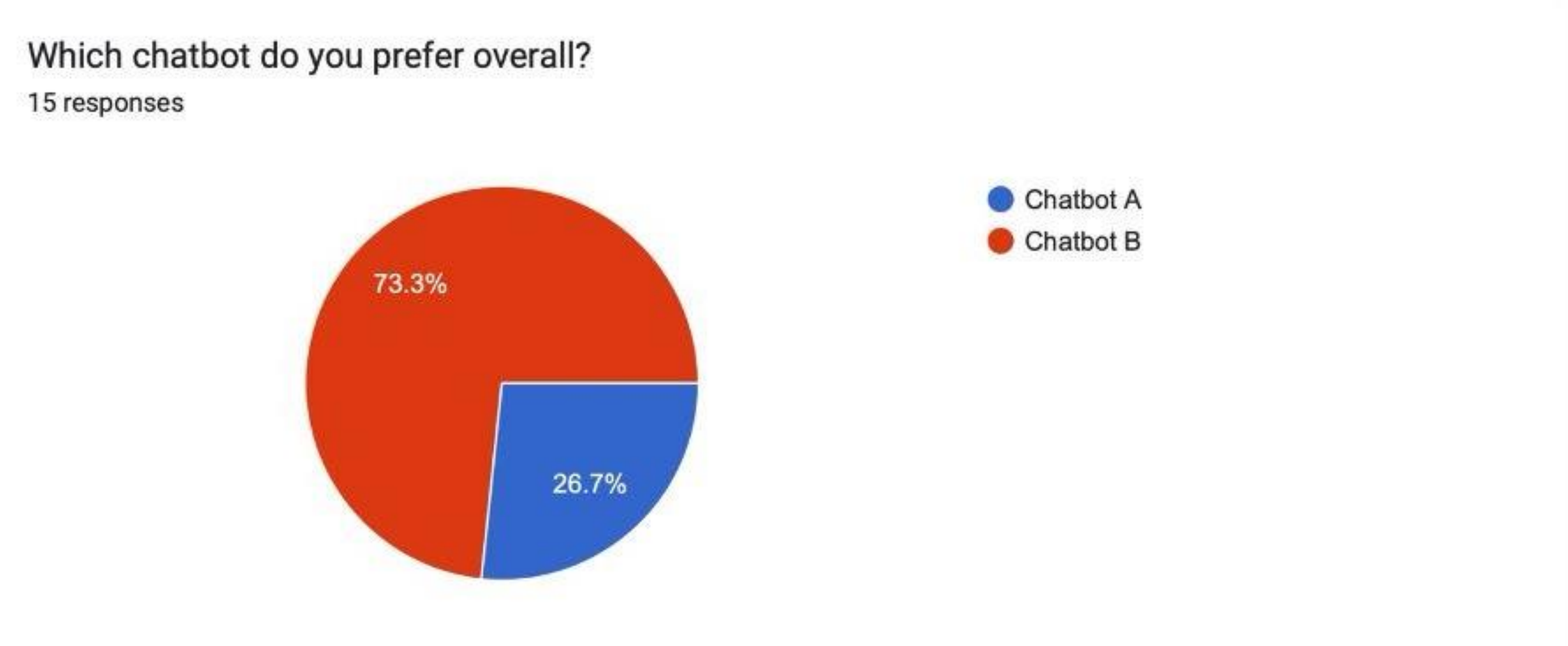
Cross Group Evaluation

	Testing Session Memory						
		Linear	Self Query	Traffic Cop	Elastic Search	NotebookLM	
	When do spring 2026 classes start?						Answer first day of spring 2026
1	When do they end?	Fails to Use Context	Correctly Uses Context	Correctly Uses Context	Fails to Use Context	Fails to Use Context	Answer last day of spring 2026 courses
	What is the hardship withdrawal policy?						Answer hardship withdrawal
2	Who approves it?	Fails to Use Context	Fails to Use Context	Fails to Use Context	Correctly Uses Context	Fails to Use Context	Provide office or contact that handles it
	How much is tuition?	Correct Clarification	Fails to use context	Correct Clarification	Correct Clarification	Fails to use context	Clarify: ask which school
	Graduate	Correctly Uses Context	Fails to Use Context	Correctly Uses Context	Correctly Uses Context	Fails to Use Context	which level
3	Never Mind	Fails to Reset Context	Correctly Resets Context	Correctly Resets Context	Fails to Reset Context	Fails to Reset Context	Clear pending, respond helpfully without answering tuition
	What is academic probation?	correct answer	correct answer	correct answer	correct answer	wrong answer	Answer academic probation
4	How long can I be on it?	Fails to Use Context	Correctly Uses Context	Correctly Uses Context	Correctly Uses Context	Fails to Use Context	Answer consecutive semester limit
	How much is tuition?	Correct Clarification	Correct Clarification	Irrelevant clarification	Correct Clarification	Fails to use context	Clarify: ask which school
5	Meis gradute						Handle Spelling errors
	When are finals?	Fails to use context	Correct Clarification	Correct Clarification	Correct Clarification	Fails to use context	ask which semester
6	Actually, what is the plagiarism policy?	Correct Topic Switch Dete…	Correct Topic Switch Dete…	Correct Topic Switch Detection	Correct Topic Switch Detection	Fails to Detect Topic Switch	Detect topic change, answer plagiarism policy
	What is the co-terminal program?						Explain co-terminal program
7	What GPA do I need to apply?	Fails to Use Context	Correctly Uses Context	Correctly Uses Context	Correctly Uses Context	Fails to Use Context	Answer co-terminal GPA requirement (3.0)
	When is the drop deadline?	Fails to Use Context	Correct Clarification	No clarification	Correct Clarification	Fails to use context	ask which semester
8	What is the academic probation policy?	Correct Topic Switch Dete…	Correct Topic Switch Dete…	Correct Topic Switch Detection	Correct Topic Switch Detection	Fails to Detect Topic Switch	Detect topic change, answer probation policy directly
	What is the maximum course load I can take each semester?	No clarification	Fails to use context	No clarification	No clarification	Fails to use context	ask which semester
9	Graduate	Incorrect Context Carryover	Incorrect Context Carryover	Incorrect Context Carryover	Incorrect Context Carryover	Incorrect Context Carryover	answer for graduate
	What is the GPA requirement for graduation?						Answer graduation GPA requirement
	What about for honors?	Fails to Use Context	Correctly Uses Context	Fails to Use Context	Correctly Uses Context	Fails to Use Context	Answer honors GPA thresholds
10	What is the highest honor?	Fails to Use Context	Correctly Uses Context	Fails to Use Context	Correctly Uses Context	Fails to Use Context	Answer honors GPA thresholds
	What is the amnesty policy?						Answer amnesty policy
11	Does it apply to graduate students too?	Fails to Use Context	Correctly Uses Context	Correctly Uses Context	Correctly Uses Context	Fails to Use Context	Answer yes/no with context

Online Evaluation

Metric		Chatbot A	Chatbot B
Avg rating	Accuracy	3.8	3.5
	Relevance	3.57	3.78
	Clarity	3.5	3.85
	Satisfaction	3.5	4.07

and students to test both the systems and provide a
on a scale of 1-5; 1 being very poor and 5 being



Conclusion and Future Work

- 4.28 / 5 mean score • 66.3% perfect answers across 89 questions
- Outperforms pure RAG on structured, entity-specific queries
- Chunk-level provenance - every answer traceable to an official IIT source

- Scheduled re-indexing pipeline for data freshness each semester
- Improved multi-topic query merging across parallel domain searches
- Expand to housing, graduate admissions, financial aid domains
- Source-linked answers - highlight exact chunk within source page

Thank you.